Different C Standards: The Story of C

This article covers the various standards of the C language, from K&R C through ANSI C, C99, C11 and Embedded C.

I have always wanted to write about the different standards of C but refrained from doing so for two reasons. The first is that though as an academician I respect and am devoted to C, I thought the industry hardly cared about it. The other more compelling reason is that not all the different standards of C are taken seriously. So, if nobody cares, readers may wonder why am I bothering to write this article.

Two incidents made me change my mind. To write an article about popular programming languages in *OSFY*, I did a detailed study of programming languages and that included C. I found out that C is still very popular in all the rankings, with supporters in both the academic world and the industry. The other reason is that in a technical interview I had a few months ago, I faced a question based on the C11 standard of C. Even though I absolutely fumbled with that question, I was quite happy and excited. Finally, the times have changed and people are now expected to know the features of C11 and not just ANSI C. So, I believe this is the right time to start preaching about the latest standards of C.

The ANSI C standard was adopted in 1989. In the last 28 years, C has progressed quite a bit and has had three more standards since ANSI C. But remember, ANSI C is not even the first C standard — K&R C holds that distinction. So, there were standards before and after ANSI C. Let's continue with a discussion of all the five different standards of C — K&R C, ANSI C, C99, C11 and Embedded C. For the purposes of our discussion, the compiler used is the *gcc* C compiler from the GNU Compiler Collection (GCC).

If there are five different standards for C, which one is the default standard of *gcc*? The answer is: none of the above. The command *info gcc* will tell you about the current default standard of *gcc*, which can also be obtained with the option *-std=gnu90*. This standard has the whole of ANSI C with some additional GNU-specific features. But there's no need to panic — *gcc* gives you the option to specify a particular standard during compilation, except in the case of K&R C. I will mention these options while discussing each standard.

But if there are so many standards for C, who is responsible for creating a new standard? Can a small group of developers propose a new standard for C? The answer is an emphatic 'No!'. The C standards committee responsible for making the decisions regarding C standardisation is called ISO/IEC JTC 1/SC 22/WG 14. It is a standardisation subcommittee of the Joint Technical Committee ISO/IEC JTC 1 of the International Organization for Standardization (ISO) and the International Electrotechnical Commission (IEC).

K&R C

A lot has been written about the history of C and I am not going to repeat it. All I want to say is that C was developed by Dennis Ritchie between 1969 and 1973 at Bell Labs. It was